

Phase 3: Project Design

EduGenie — Google Gemini Powered Learning Assistant

1. Executive Summary

EduGenie is a lightweight AI-powered educational assistant designed to streamline and democratize learning through Generative AI. Built with FastAPI for backend logic, Jinja2 template rendering, HTML/CSS frontend, and dual AI model pipelines (Google Gemini 1.5 Pro and LaMini-Flan-T5-783M), EduGenie provides smart Q&A, detailed explanations, quiz generation, text summarization, and adaptive learning paths.

2. High-Level Architecture

EduGenie uses a client-server architecture designed for cloud intelligence and lightweight local execution:

- **Frontend Layer:** HTML5 forms (`index.html`), CSS styling (`style.css`), and JavaScript asynchronous fetch calls.
 - **Backend API Layer:** FastAPI ASGI framework (`main.py`) powered by Uvicorn.
 - **AI / LLM Integration Layer:**
 - **Gemini 1.5 Pro (Cloud API):** Handles multi-turn Q&A (`/qa`), JSON Quiz Generation (`/quiz`), Passages Summarization (`/summarize`), and Adaptive Learning Recommendations (`/learn/recommendations`).
 - **LaMini-Flan-T5-783M (Local HuggingFace Transformers):** Executes lightweight concept explanations (`/explain`) locally on CPU/Apple Silicon devices.
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3. System Components & File Structure

EduGenie/

- └─ Phase-3-Project-Design/
 - | └─ System_Architecture.png
 - | └─ Use_Case_Diagram.png
 - | └─ Project_Design.docx
- └─ app/
 - | └─ main.py # FastAPI main entrypoint & REST endpoints
 - | └─ explanation_module.py # LaMini-Flan-T5 model wrapper
 - | └─ qna.py # Gemini 1.5 Pro Q&A logic
 - | └─ quiz_module.py # Gemini 1.5 Pro JSON Quiz generator
 - | └─ summary_module.py # Text summarization pipeline
 - | └─ learning_path.py # Adaptive learning path engine
 - | └─ templates/
 - | └─ index.html # User interface templates
 - | └─ static/
 - | └─ style.css # UI styling
- └─ requirements.txt # Project dependencies
- └─ README.md # Documentation

4. Detailed Module Design & Endpoints

Endpoint	Method	Underlying Model	Primary Logic & Objective
/explain	POST	LaMini-Flan-T5-783M (L...	Loads local tokenizer and seq2seq model to simplify

Endpoint	Method	Underlying Model	Primary Logic & Objective
			complex educational concepts specifically tailored for school students or beginners. Key generation parameters: <code>max_new_tokens=150</code> , <code>temperature=0.7</code> , <code>top_k=50</code> , <code>top_p=0.95</code> .
/qa	GET ▾	Google Gemini 1.5 Pro ▾	Takes a user query parameter (<code>question</code>) and returns concise, accurate, and contextually aware answers across academic subjects.
/quiz	POST ▾	Google Gemini 1.5 Pro ▾	Accepts passage or topic text, prompts Gemini to construct 3 multiple-choice questions (MCQs) with 4 choices and the exact correct answer in JSON format, cleans Markdown code block fences with regex (<code>clean_json_block</code>), and returns parsed JSON.
/summarize	POST ▾	Google Gemini 1.5 Pro ▾	Condenses long educational articles, history notes, or scientific texts into digestible summaries while maintaining core information.
/learn/recommendations	GET ▾	Google Gemini 1.5 Pro ▾	Generates adaptive learning roadmaps spanning Beginner (1-2 weeks), Intermediate (2-3 weeks), and Advanced levels (3-4 weeks), complete with key topics, order of learning, online resources, and books.

4.1 Explanation Module (/explain)

- **Model:** MBZUAI/LaMini-Flan-T5-783M (Hugging Face Transformers / PyTorch)
- **Method:** POST
- **Logic:** Loads local tokenizer and seq2seq model to simplify complex educational concepts specifically tailored for school students or beginners with parameters `max_new_tokens=150, temperature=0.7, top_k=50, top_p=0.95`.

4.2 Q&A Module (/qa)

- **Model:** Google Gemini 1.5 Pro
- **Method:** GET
- **Logic:** Takes a user query parameter (`question`) and returns concise, accurate, and contextually aware answers across academic subjects.

4.3 Quiz Generation Module (/quiz)

- **Model:** Google Gemini 1.5 Pro
- **Method:** POST
- **Logic:** Accepts passage or topic text, prompts Gemini to construct 3 multiple-choice questions (MCQs) with 4 choices and the exact correct answer in JSON format, cleans Markdown code block fences with regex (`clean_json_block`), and returns parsed JSON.

4.4 Summarization Module (/summarize)

- **Model:** Google Gemini 1.5 Pro
- **Method:** POST
- **Logic:** Condenses long educational articles, history notes, or scientific texts into digestible summaries while maintaining core information.

4.5 Learning Path Module (/learn/recommendations)

- **Model:** Google Gemini 1.5 Pro
 - **Method:** GET
 - **Logic:** Generates adaptive learning roadmaps spanning Beginner (1-2 weeks), Intermediate (2-3 weeks), and Advanced levels (3-4 weeks), complete with key topics, order of learning, online resources, and books.
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5. Security & Deployment Best Practices

- **Environment Management:** API keys (e.g., `GEMINI_API_KEY`) must be stored in `.env` files and added to `.gitignore`. Never commit raw API keys to repository.
- **Local Server Running:** Execute `uvicorn main:app --reload` on host `127.0.0.1:8000`.